

Project Bob: Multi-Source Threat Fusion Engine

Continuous Cyber & Intel Correlation, MITRE ATT&CK Mapping,
and Actionable BLUF Generation

Team Neural-Nomads | Defense Threat Intelligence Track

The Operational Challenge

- **Data Overload**

Defense teams receive thousands of alerts from SIEM systems, satellite telemetry, cyber sensors, and intelligence reports.

- **Format Fragmentation**

Structured logs, unstructured text, and other feed formats make manual analysis slow and inconsistent.

- **False-Positive Pressure**

Analysts can waste critical time investigating harmless events while genuine threats compete for attention.

- **Decision Velocity**

Commanders need a concise, prioritized picture in minutes rather than raw alert dumps.

What We Are Building

- **Unified Ingestion**

Bring multi-source alert data into one normalized representation so downstream processing follows a common structure.

- **Correlation**

Group related alerts and compare signals across sources to identify patterns instead of treating every alert independently.

- **Prioritization**

Apply transparent scoring so analysts can focus first on the most important and highest-confidence events.

- **MITRE ATT&CK Mapping**

Associate observed attacker behavior with the MITRE ATT&CK framework to give analysts a standardized view of techniques.

- **BLUF Generation**

Turn the analysis into a Bottom Line Up Front summary that presents the key finding, priority, evidence, and impact.

End-to-End Workflow

- **1. Ingest**

Accept representative SIEM, cyber-sensor, telemetry, and report data.

- **2. Normalize**

Convert different source formats into a consistent internal alert structure.

- **3. Correlate**

Connect related events using common fields and time/context relationships.

- **4. Score**

Calculate a priority/risk score using severity, source confidence, correlation, and recurrence.

- **5. Map**

Identify relevant MITRE ATT&CK techniques from the observed behavior.

- **6. Generate BLUF**

Use watsonx.ai to turn structured analysis into a concise commander-facing summary.

- **7. Display**

Render prioritized alerts, ATT&CK mappings, and BLUF cards in the dashboard.

System Architecture



IBM Bob / MCP / watsonx.ai integration

The architecture separates data processing from AI-generated language output. Analytical functions can be exposed through the MCP layer, while watsonx.ai is used for the natural-language BLUF stage.

Target flow: User → IBM Bob CLI → MCP Server → analytical tools / data → watsonx.ai → dashboard

MITRE ATT&CK & Noise Reduction

- **Behavior Mapping**

Observed alert behavior is mapped to ATT&CK techniques to provide a consistent analytical vocabulary.

- **Cross-Source Correlation**

Related events are evaluated together instead of relying on a single noisy signal.

- **Priority Scoring**

Threat priority combines factors such as severity, confidence, asset importance, correlation, and recurrence.

- **Explainable Results**

The dashboard should show why an alert received its priority so analysts can review the evidence.

AI + IBM Technology Integration

- **IBM Bob**

Used as the development/agent workflow layer for building and working with the project.

- **MCP Server**

Acts as the integration layer that exposes analytical capabilities to the AI workflow.

- **watsonx.ai**

Generates the natural-language BLUF output from structured analytical results.

- **Backend Analytics**

Processes and scores representative alert data before the AI summary is produced.

- **Dashboard**

Presents the resulting priority, evidence, ATT&CK mapping, and BLUF in a usable interface.

Commander BLUF — Example Output

Priority Level:

CRITICAL (P1)

BLUF:

Hostile actor initiated command-and-control communication across SATCOMLink-4, correlated with anomalous telemetry at Sector 7. High probability of operational network compromise.

ATT&CK TTPs:

T1071 (Application Layer Protocol), T1059 (Command and Scripting Interpreter)

Impact:

Potential disruption to tactical relay communications within 60 minutes.

Recommended Action:

Isolate SATCOM Gateway 4-B, apply secondary encryption keys, deploy Cyber Protection Team 2.

MVP Scope & Demonstration Plan

- **Synthetic Data First**

Use representative/sample alerts rather than real classified, sensitive, or operational feeds.

- **Core Demo**

Upload or load sample alerts → normalize → correlate → score → map → generate BLUF → display dashboard.

- **Visible Evidence**

Show the input, processing result, priority score, ATT&CK mapping, and generated BLUF during the demo.

- **Working Over Ambitious**

Prioritize a reliable end-to-end prototype over unsupported claims of real-time military deployment.

- **Demo Package**

Provide a 3–5 minute video, at least three sequential screenshots, and the required presentation PDF.

Team Execution & Deliverables

- **Yagna — Team Representative**

Repository coordination, submission metadata, architecture, integration, MCP/server coordination, and final validation.

- **Dhyey — Backend & Data Core**

Backend/data processing, database or data structures, API/data pipeline work, environment example, and setup guide.

- **Het — D2 Problem Lead + UI/Media**

D2 threat engine, prioritization, MITRE/BLUF flow, dashboard, presentation, screenshots, and demo video.

- **Sarim — Documentation & Support**

README formatting, spelling/grammar, placeholder checks, and documentation proofreading.

Expected Demo Story

- **Problem**

Start with the analyst's alert overload and the cost of false positives.

- **Input**

Show several representative alerts arriving from different sources.

- **Fusion**

Demonstrate normalization and correlation of related events.

- **Decision Support**

Show the priority score and MITRE ATT&CK mapping.

- **AI Output**

Generate and display the BLUF summary through the AI integration.

- **Impact**

Finish by showing how the workflow gives analysts a faster, structured starting point for investigation.